# Development of an IoT-Based Smart Agricultural System for Environmental Monitoring and Soil Condition Using Raspberry Pi and Mobile Interface

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***Abstract*— The growing global population, expected to reach 9.8 billion by 2050, places immense pressure on food production and highlights the urgency of addressing food security and crop wastage. Traditional agricultural practices, which rely heavily on manual observation and farmer intuition, are increasingly insufficient due to labor shortages, inefficiency, and inconsistency in monitoring accuracy. This paper presents the development of an IoT-based smart agricultural system for real-time environmental and soil condition monitoring to support precision farming. The objective is to develop an IoT based smart agricultural system for environmental monitoring and soil condition using Raspberry Pi and mobile interface. The system integrates a plant monitoring chamber with DHT22 sensors for temperature and humidity, TSL2591 for light intensity, and YL69 for soil moisture. Sensor data is collected via Raspberry Pi units and transmitted to Blynk Cloud to be displayed through a mobile application and web interface. The result shows consistent readings of four sensor parameters over a two-day measurement. This approach aims to reduce crop losses, optimize growing conditions, and enhance overall agricultural productivity, offering a scalable solution especially beneficial for small-scale and remote farming operations.**

***Keywords—environmental sensors, IoT monitoring system, precision farming, Raspberry Pi, smart agriculture***

## I. INTRODUCTION

Agriculture has long been vital to human survival and remains a key global industry, employing approximate a third of the world's workforce. With the global population steadily rising, the demand for food continues to increase. The United Nations projects that the global population will reach 9.8 billion by 2050 [1], which will require a 70% increase in food production to meet the demand. This growing pressure is compounded by challenges such as climate change, soil degradation, water scarcity, and crop diseases. Pests and diseases alone are estimated to cause 20–40% of global crop losses annually, according to the Food and Agriculture Organization (FAO).

Traditional plant growth monitoring is primarily conducted through manual visual observation, which is labour-intensive, time-consuming, and prone to human error [2]. This approach is no longer sufficient given the increasing challenges faced by modern agriculture. At the same time, farmers are confronted with worsening environmental conditions such as extreme temperatures, soil degradation, droughts, and increasingly unpredictable monsoon rainfall [1]. These factors hinder effective crop management and irrigation planning, further exacerbating food insecurity. In addition, overexploitation of soil resources has accelerated land degradation, threatening long-term agricultural sustainability.

To address the rising food demand driven by rapid population growth, the adoption of more sustainable agricultural practices has become increasingly critical. The emergence of the Internet of Things (IoT) offers a promising solution for modernizing agriculture through smart farming technologies. By integrating sensors and actuators across fields and equipment, farmers can collect valuable real-time data on key parameters such as temperature, soil moisture, water usage, and fertilizer application. This data-driven approach enables more precise and efficient farm management. Consequently, implementing an IoT-based plant growth monitoring system can significantly enhance decision-making, reduce resource wastage, and improve overall crop productivity, making it a practical and impactful tool for today's farmers and agriculturists.

Previous work reported the monitoring of the condition of banana leaf by combining IoT and image processing to improve product yield [2]. However, image processing technique requires high resolution cameras that will cause the cost to increase. Selection of microcontroller is crucial for the development of an IoT-based smart agricultural system for environmental monitoring and soil condition. Hence, in this paper, we proposed the development of an IoT-based plant monitoring system using Raspberry Pi as a practical alternative to conventional methods, aiming to reduce crop wastage, improve decision-making, and support sustainable agriculture.

## II. IOT IMPLEMENTATION IN AGRICULTURE

Technological advancements and shifting agricultural practices have continuously shaped the evolution of farming. Over a third of the world's workforce is employed in agriculture [3]. A country like India is an agriculture-based country where agriculture is not only a source of income but also a way of life. Agriculture supports two-thirds of the population, either directly or indirectly [2]. Moreover, the agriculture sector contributes significantly to global economic growth, including in Malaysia whereby the contribution of agriculture to the overall Gross Domestic Sector (GDP) was 7.4% in 2020 [4]. While industrialized countries have increased agricultural productivity through advanced technologies and modern farming methods which thereby reduce the reliance on manual labor, many developing nations are still working to realize their full agricultural potential [3].

The Internet of Things (IoT) has emerged as a transformative technology in the field of agriculture, commonly referred to as smart farming. By connecting physical devices such as sensors, actuators, and communication modules, IoT enables real-time monitoring and automation of various agricultural processes. This technology allows farmers to gather precise data on environmental conditions such as temperature, humidity, soil moisture, and light intensity which are critical for optimizing crop growth and yield. With the ability to remotely track these parameters, farmers can make timely and informed decisions, ultimately improving efficiency, reducing labour dependency, and minimizing crop losses.

One of the key benefits of IoT in agriculture is precision farming. Instead of relying on manual observation or fixed schedules, IoT systems enable targeted interventions based on actual field conditions. For example, irrigation systems can be automated to deliver water only when and where it is needed, reducing water waste and improving plant health. Similarly, fertilizers and pesticides can be applied more effectively based on sensor data, lowering costs and minimizing environmental impact [5]. IoT also supports livestock management, greenhouse automation, and crop health monitoring using data-driven insights.

There are some previous studies utilize image processing to determine the health and the disease of the plant. For example in [2], the researchers implemented image processing to monitor the condition of banana leaf and paddy with the objective to improve product yield and reduce crop wastage. The proposed system comprises of Arduino UNO microcontroller, temperature and humidity sensors, and a camera. Another related work reported in [6] focuses on greenhouse automation and monitoring system whereby the sensors used consist of temperature and humidity sensor (DHT11), and soil moisture sensor (YL69). The system collect data in a greenhouse through the Raspberry Pi 3 Model B and sent to IoT cloud which is connected to the end user web application. The system proposed in [3] focuses on a user-friendly mobile interface for plant growth monitoring. It uses DHT11 and SHT11 sensors to measure temperature, humidity, and soil moisture. A Raspberry Pi is used as the main controller, and BLE technology transmits data to the user's phone. The system also includes a height measuring device to track plant growth. Unlike the system in [6], which uses a web-based interface, this project relies on a mobile application. . In this paper, we reported the development of an IoT-based plant monitoring system using Raspberry Pi together with Blynk interface.

## III. Methodology

This section outlines the framework of the proposed environmental and soil condition monitoring system.

### A. *Overview on the Proposed Monitoring System*

Fig. 1 illustrates overall system block diagram for plant monitoring system. The selected components of the system includes Raspberry Pi 4 Model B, TSL2591 for the light sensor, DHT22 for the temperature and humidity sensor, YL69 for the soil moisture sensor. The proposed monitoring system uses a monitoring chamber for maintaining constant growing conditions to eliminate variables that could skew the results of the experiment. The data collected by the sensors were sent and uploaded to a cloud storage service which is Blynk Cloud via Wi-Fi. Then, the uploaded data would be displayed on a mobile application called Blynk. Finally, the user can access the monitoring data whenever Wi-Fi connection is available. Fig. 2 shows a schematic diagram of the connections between the sessors and Raspberry Pi.

### B. *Experimental Setup*

The monitoring chamber was made of a used bucket as shown in Fig. 3. The monitoring chamber was wrapped with a transparent plastic so that light can pass through. It has a base dimension of 350 mm by 350 mm and a height of 320 mm. This dimension is based on the observation made on water spinach whereby the plant has a width and length of 200 mm by 200 mm, and height of approximately 200mm. The plant was placed in the middle of the chamber. Several openings were cut around the sides and top of the bucket using a hand grinder to allow proper sensor placement and airflow. The light intensity sensor (TSL2591) was positioned at the top of the monitoring chamber to ensure an unobstructed view, allowing it to accurately measure ambient light levels. The relative humidity and atmospheric temperature sensor (DHT22) is placed beside the water spinach plant inside the chamber to closely reflect the surrounding environmental conditions. Meanwhile, the soil moisture sensor (YL69) is inserted directly into the plant's soil to monitor moisture levels effectively. Raspberry Pi which is responsible for interfacing with all sensors and handling data transmission, was located outside the chamber in an accessible position. This placement ensures ease of maintenance while keeping the internal environment of the monitoring system undisturbed.

The monitoring chamber is placed indoors to allow easier control of environmental variables such as light intensity and temperature. This setup enables convenient adjustments by simply relocating the chamber to areas with different conditions, making the system more flexible for experimentation and monitoring.

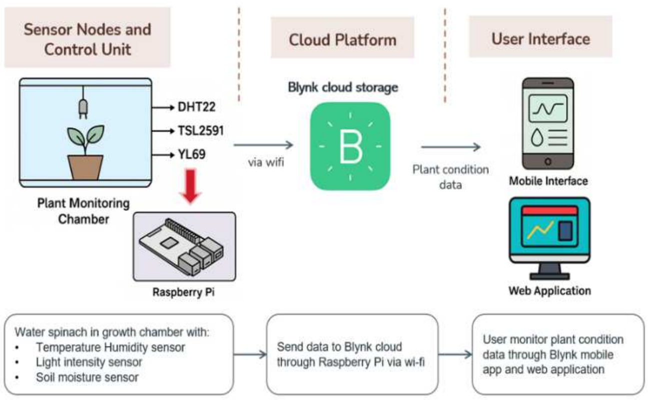


Fig. 1. Proposed System Block Diagram

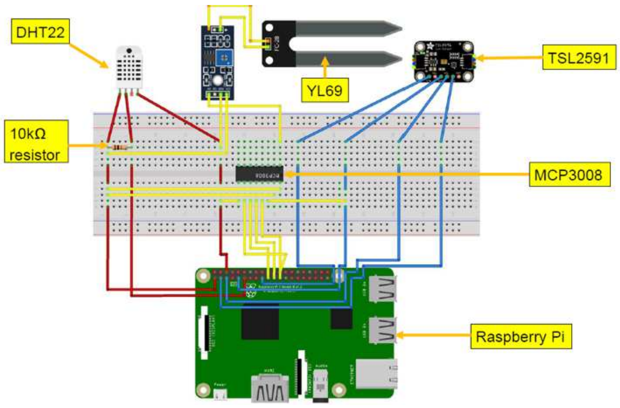


Fig. 2. Circuit Connection of the System


Fig. 3. Water Spinach Monitoring Chamber

As shown in Fig. 5, the plant pot was placed inside a container filled with water, eliminating the need for regular watering. This approach not only ensures consistent soil moisture but also prevents the sensors from getting wet during manual watering. Additionally, the presence of standing water within the enclosed monitoring chamber helps maintain high humidity levels, as the evaporated water remains trapped in the chamber. This setup effectively creates a stable microclimate that supports healthy plant growth while protecting the sensors.

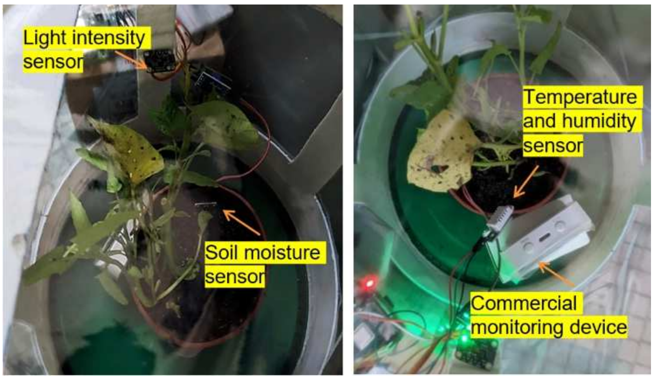


Fig. 4. Sensors Placement in Monitoring Chamber

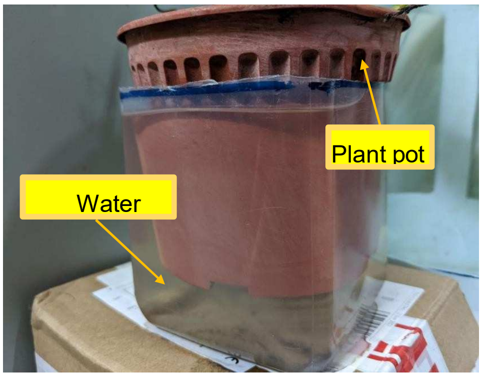


Fig. 5. Plant Pot inside a Container of Water

## IV. Results and Discussion

### A. *Prototype of Plant Monitoring System*

The plant monitoring system prototype was placed indoor inside the dorm room so that variables could be adjusted easily. The water spinach was placed inside the monitoring chamber to keep the humidity high. The commercial monitoring device, which is the white device with a screen, was also placed inside the monitoring chamber just beside the plant as shown in Fig. 3.

The TSL2591 light intensity sensor was placed on one of the support or pillars of the chamber while the YL69 soil moisture sensor is placed fully inside the soil. Fig. 4 shows a clearer picture of the placement of both TSL2591 and YL69 sensor inside the monitoring chamber. The light intensity sensor cannot be placed too high on the support of the chamber since light can be blocked from being detected by the top of the chamber. The light intensity sensor was also placed 90° of the floor so that it can detect light from a wider range and more accurate direction. Next, the DHT22 temperature-humidity sensor was placed just beside the plant to get the most accurate reading as shown in Fig. 4.

### B. *System Accuracy Testing*

The sensors for the plant monitoring system was first tested for their accuracy and reliability. This is to ensure that the monitoring of environmental and soil condition of a plant inside the chamber is working as intended. The sensors accuracy was measured by recording the data every 1 minute for 30 minutes on an evening and comparing them with a commercial sensor by Tuya. The commercial device can detect three parameters which are temperature, humidity, and light intensity. Table I presents comparison of the average sensors reading in 30 minutes between sensors used in the proposed system and commercial device. The commercial device used for reference showed completely constant readings for both humidity and light intensity throughout the measurement period. This indicates that the sensors in the commercial device may have lower sensitivity or slower response to minor environmental changes. In contrast, the sensors used in the developed IoT-based system; particularly the DHT22 and TSL2591, were able to detect and reflect small fluctuations in environmental conditions over time. This suggests that the sensors in the proposed system are more responsive and capable of capturing real-time variations, which is crucial for accurate plant growth monitoring. Higher sensitivity allows for better detection of microclimatic changes, enabling timely intervention and improved decision-making in precision agriculture. The small percentage error observed between the two sets of readings suggests that the sensors in the proposed system are not only responsive but also accurate. This minimal deviation supports the validity of the sensor outputs, indicating that they are capable of producing reliable real-time data.

TABLE I. COMPARISON OF SENSORS READINGS FROM PROPOSED SYSTEM AND COMMERCIAL DEVICE

| Parameters | Average Readings | | Percentage error (%) |
|---|---|---|---|
| | Proposed sensors | Commercial Device | |
| Temperature | 30.2 | 29.8 | 1.34 |
| Humidity (%) | 92.6 | 91.4 | 1.31 |
| Light Intensity (Lux) | 13.5 | 13 | 3.8 |
| Soil Moisture (%) | 69.7 | N/A | N/A |

### C. System Reliability Testing

Table II presents the sensor readings collected at three time intervals: morning (0800 hrs), noon (1200 hrs), and evening (1800 hrs), over a two-day period to assess the reliability of the sensors used. The plant was placed indoors in a dormitory room, with no direct exposure to sunlight. The data for light intensity, measured using the TSL2591 sensor, shows consistent and comparable values at corresponding times on both days. On the first day, the light intensity readings were 9.4 lux (0800 hrs), 38.3 lux (1200 hrs), and 10.7 lux (1800 hrs). On the second day, the readings were 8.0 lux (0800 hrs), 31.3 lux (1200 hrs), and 11.6 lux (1800 hrs). A clear pattern is observed, where light intensity was lowest in the morning and evening and peaked around noon. This consistent trend across both days suggests that the TSL2591 sensor is reliable for measuring light intensity.

In contrast, the temperature readings obtained using the DHT22 sensor showed a gradual increase throughout the day. On the first day, temperatures recorded were 29.2°C (0800 hrs), 29.3°C (1200 hrs), and 30.2°C (1800 hrs). The second day showed a similar increasing trend, with temperatures of 29.2°C, 29.5°C, and 29.7°C at the same respective time points. Although the changes were relatively small, the repeated pattern across both days confirms the consistency of the sensor. Therefore, the DHT22 sensor is deemed reliable for measuring ambient temperature.

For humidity readings, no consistent trend was observed across both days. On the first day, humidity decreased from 90.2% (0800 hrs) to 88.7% (1200 hrs), then slightly increased to 89.8% (1800 hrs). On the second day, a steady decrease was recorded from 93.4% to 89.3%. These variations could be due to multiple environmental factors affecting humidity levels during the day. However, the values on both days remained close and within a reasonable range, indicating that the DHT22 sensor provides consistent and reliable humidity measurements.

For soil moisture, only slight changes were observed in the readings. On the first day, values ranged from 95.8% to 96.0%, while on the second day, they varied between 96.1% and 96.4%. These small differences were expected, as the plant pot was placed in a container filled with water, maintaining high and stable soil moisture levels. The consistency in the readings supports the reliability of the soil moisture sensor.

TABLE II. RELIABILITY TESTING OF MONITORING SENSORS

| Day | Time (Hrs) | Sensors reading | | | |
|---|---|---|---|---|---|
| | | Light Intensity (lux) | Temperature (°C) | Humidity (%) | Soil Moisture (%) |
| Day 1 | 0800 | 9.4 | 29.2 | 90.2 | 95.8 |
| | 1200 | 38.3 | 29.3 | 88.7 | 95.2 |
| | 1800 | 10.7 | 30.2 | 89.8 | 96.0 |
| Day 2 | 0800 | 8.0 | 29.2 | 93.4 | 96.1 |
| | 1200 | 31.3 | 29.5 | 91.8 | 95.8 |
| | 1800 | 11.6 | 29.7 | 89.3 | 96.4 |

### D. User Interface

The Blynk mobile application was successfully integrated with the monitoring system, allowing real-time visualization of data from the DHT22, TSL2591, and YL69 sensors. As illustrated in Fig. 6, the interface displays four key parameters: temperature, humidity, light intensity, and soil moisture. Temperature and humidity readings are shown using gauge widgets, with ranges set from -40°C to 80°C and 0% to 100%, respectively. Light intensity and soil moisture values are displayed using labelled value widgets, with units in lux and percentage. The mobile interface was intentionally designed to be simple and user-friendly, enabling users to quickly interpret sensor data at a glance without requiring technical knowledge.

The Blynk interface is also accessible via a web dashboard, as shown in Fig. 7. This dashboard provides users with a more detailed view of the plant monitoring data, including interactive graphs for humidity, temperature, and soil moisture. These graphs display sensor readings over time, allowing users to observe trends and fluctuations in environmental conditions. Additionally, the web interface offers flexible data viewing options—users can choose to view live data, data from the past hour, the last 6 hours, or define a custom date range for analysis. This extended functionality enhances the monitoring experience, enabling more informed decision-making based on historical and real-time data.

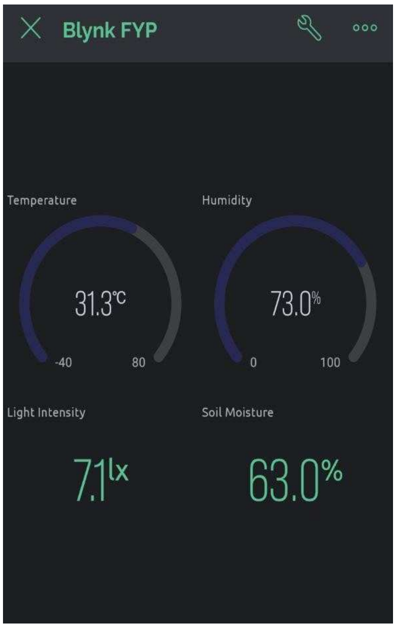


Fig. 6. User Interface of Blynk Mobile Application


Fig. 7. Web interface of Blynk application .

## V. Conclusion

In response to the growing global population and rising food demand, the adoption of more sustainable and efficient agricultural practices is essential. This project demonstrates that an IoT-based plant monitoring system using Raspberry Pi offers an affordable alternative to expensive commercial solutions. By integrating environmental sensors and enabling remote access through the Blynk mobile application, the system enhances the efficiency of plant monitoring and supports timely, data-driven decision-making for farmers. However, before such systems can fully replace traditional visual observation methods, they must undergo thorough testing to ensure accuracy, reliability, and long-term usability in real farming conditions. With proper validation and refinement, this approach has strong potential to contribute to smarter, more productive, and sustainable agricultural practices.

Several enhancements can be made to improve the current prototype of the plant monitoring system. One of the key features that could be added is an alert notification mechanism. This feature would notify users when any monitored parameter such as temperature, humidity, light intensity, or soil moisture; exceeds a predefined threshold. Such real-time alerts would enable users to take immediate corrective actions, such as relocating the plant to a better environment or watering it when soil moisture is low. Another significant improvement is the integration of an automated watering system. By connecting a water pump to the system, the plant could be watered either at scheduled intervals or automatically whenever the soil moisture level drops below a certain limit. This automation would be especially beneficial for users who are frequently occupied with work or travel, as it reduces the need for constant manual supervision. In the future, the system could also incorporate machine learning algorithms to analyse long-term data and provide insights or recommendations for optimal plant growth. This would elevate the system from simple monitoring to intelligent decision support, further enhancing its value in smart agriculture applications.


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